

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We pre-registered and ran a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D (1D-ResNet-34)), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (the operating CI is BCa at $B = 10,000$, fully re-run on the entire 60-experiment grid; earlier percentile CIs at $B = 10,000$ in 31/60 experiments and $B = 200$ in 21/60 are retained as historical provenance, and the two CI methods agree in sign in 60/60 experiments). The pre-registered primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a registered empirical direction prior. Across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced), TS yields a positive OOD

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benefit in 51/60 cases (85.0%): InceptionTime 27/30 (90.0%), ResNet1D 24/30 (80.0%); nine counter-examples are transparently disclosed with post-hoc mechanistic interpretations. When six experiments with degenerate CI widths ($< 3 \times 10^{-4}$) are excluded, the headline count becomes 45/60 (75.0%), so the 85.0% rate should be read jointly with effect sizes and CI diagnostics. The pre-registered ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met ($23/60 = 38.3\%$ under the BCa operating CI; the earlier percentile CIs gave $29/60 = 48.3\%$), triggering the pre-registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (means $+0.0159$ vs. $+0.0083$) and cannot be attributed exclusively to OOD-specific mismatch. A 27-cell Shapley decomposition is validated synthetically ($n = 20,000$), a method-selection boundary separates TS from the weakest methods (EM prior and Matrix scaling), and the predictability experiment (LOO R^2 CI upper bound $0.104 < 0.5$) triggers the pre-registered failure branch. A discrimination-aware calibration gate (C5) decouples calibration effectiveness from clinical deployability: 7/14 pair $\times$ architecture strata pass both calibration and discrimination layers. All hypotheses are tagged *exploratory + robustness validation*, not confirmatory. In summary, we contribute (i) the first multi-seed quantification of OOD calibration benefit with pre-registered endpoint; (ii) the ID boundary condition analysis with pre-registered branch triggered; (iii) Shapley decomposition validation on 27 synthetic cells; (iv) the method-selection boundary separating TS from fragile prior-correction methods; and (v) a discrimination-aware deployment gate reframing the 9 counter-examples into calibration vs. discrimination failures.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

Why this matters for ECG. PTB-XL [3], Chapman-Shaoxing [4], and CPSC differ in acquisition protocols, lead sets, and label vocabularies. A model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [23]. Clinicians and regulators need a decision rule: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling and pre-registration. Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target the source-fit temperature is mismatched to the target logits and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing,

protocol v2.1-A1 [24] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from **decay** to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [25, 26]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported transparently.

Contributions. Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID

point estimates are positive and $2/60 = 3.3\%$ have $\Delta ECE_{ID} < -0.01$ (counter-example 6, -0.0152 ; one marginal case, -0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$) recovers ground truth in the identifiable regime (27/27 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is fragile when ill-conditioned cells ($w_{ID} > 10$) are included (InceptionTime 15/30); after exclusion the EM OOD finding attenuates. The mechanism is prior-likelihood mismatch: EM fits a target prior that amplifies shift-induced miscalibration when the source confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta ECE_{OOD} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 14 pair $\times$ architecture strata, 7/14 pass both layers; 7/14 fail the discrimination layer, including directions with positive ΔECE_{OOD} but accuracy below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta ECE = +0.029$, $acc = 0.657 < 0.713$). The gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

Prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [6] proposed the static calibration error, and Kumar et al. [16] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [7] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [14, 15] established a calibration hierarchy from utopia to empirical data and argued that calibration is the “Achilles heel” of predictive analytics. Strodtzoff et al. [17] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

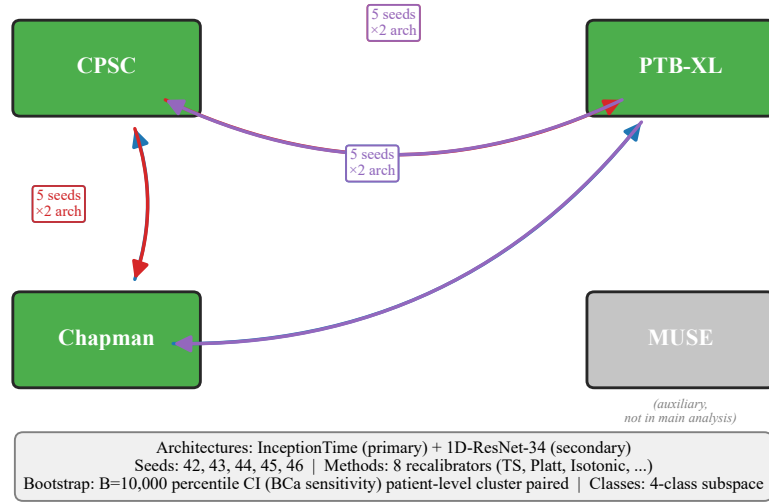


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

2.2. Prior-Shift Estimation and Transfer Calibration

Lipton et al. [9] introduced black-box shift estimation (BBSE) for detecting and correcting label shift using confusion-matrix inverses. Alexandari et al. [8] showed that maximum-likelihood calibration with bias correction is hard-to-beat at label shift adaptation, providing the theoretical foundation for our EM prior-shift method. For transfer calibration upper bounds, TransCal [20] estimates transferability for auto-selecting target-domain unsupervised optimal T ; PseudoCal [21] uses pseudo-labels for calibrated transfer; and LaSCal [22] provides label-aware shift calibration with distribution-free guarantees. Our S1 paradigm differs fundamentally: we fit on the *source* calibration split only (no target labels), with S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly weaker assumption than TransCal/PseudoCal, which access target logits or labels.

2.3. Cross-Domain ECG Calibration

Barandas et al. [10] evaluated uncertainty quantification methods in multi-label ECG classification, finding that ensemble-based methods outperform single-model approaches under subpopulation shift. Vranken et al. [11] studied uncertainty estimation for 12-lead ECG deep learning and reported significant calibration drift across acquisition devices. Haekal et al. [18] performed cross-dataset validation for atrial fibrillation detection with calibration-aware probability analysis, and Patel & Beedala [19] documented calibration drift under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general vision [1]; we contribute the multi-seed ECG quantification and the boundary-condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

2.4. ECG Foundation Models and Transfer Learning (2024–2026)

Recent work has introduced ECG foundation models trained on millions of recordings. ECGFounder [27, 35] (10 million ECGs, Harvard–Emory) reaches expert-level performance on 80 diagnoses and transfers cross-domain via fine-tuning. ECG-FM [28] is an open-weight transformer pretrained on 1.4 million ECG segments with hybrid contrastive and generative self-supervision, showing strong cross-dataset generalizability and label efficiency. HeartLang [29] treats heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked sentence pre-training. Tan et al. [30] found that fine-tuning benefits diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD [31] quantifies distribution-shift severity via uncertainty estimation to guide selective LoRA layer unfreezing, and BeatRhythm-TTA [32] performs test-time adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al. [33] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source generalization, and Chen et al. [34] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer but not the *calibration* of transferred models, the gap our study fills via pre-registered cross-corpus evaluation of recalibration methods.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptb-xl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n = 11$) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. Label Mapping and Leakage Guard

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. Methods

4.1. Transfer pairs and shift matrix

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC, all recomputed on the 4-class subspace. The L2 eval-time shift matrix (no retraining) covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise injection (BWL/MA/EM) at SNR $\{24, 12, 6, 0, -6\}$ dB; gain $\times 0.5 / \times 2$.

4.2. Architecture and seeds

We use two architectures: InceptionTime (primary, 46,000–184,000 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The BiMamba architecture is included only as a toy experiment ($n = 60$ records, downsampled resolution) and deferred to the supplement; the full-resolution BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the main-endpoint analysis or the architecture-robustness claim. Training: AdamW + cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training; temperature applied only as post-hoc temperature scaling.

4.3. Calibration methods and scenarios

We compare the no-recalibration baseline (None) against eight recalibration methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method

pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4); S1' denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

One-sided vs. two-sided choice. The one-sided alternative $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$ follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit temperature yields, empirically, little calibration change, whereas on a shifted target it is mismatched and a non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a theorem: TS applied to an under-confident target can increase ECE, and our own data contain nine negative cells. The choice was made before the pilot results, which only triggered the endpoint re-location (revision A1). The one-sided test is the pre-registered choice, archived before multi-seed unblinding.

Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [13]; stratified pooling: strata = transfer pair $\times$ architecture.

CI method note (BCa vs. percentile). The pre-registered CI method is BCa [13], at $B = 10,000$, re-run on the entire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain as historical provenance (5 lack log provenance, see Limitations); percentile CIs serve only as a sensitivity cross-check.

Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle), decay = $\Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$ 13 shift levels = 156 primary comparisons; the multi-seed validation is robustness evidence (60 seed experiments), not a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from `decay` to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

Pre-registration revision A2 (family reduction). The original BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md`.

4.6. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is

checked via the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.7. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.8. Net Clinical Value (NCV)

Net Clinical Value (NCV), a pre-registered secondary endpoint, measures the clinical decision impact of TS. For each (sample, class) pair we compare the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

5. Results

5.1. Primary endpoint: $\Delta\text{ECE}_{\text{OOD}}$ multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments ($6 \text{ pairs} \times 2 \text{ archi-}$

tectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta\text{ECE}_{\text{OOD}} = 0$ vs $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

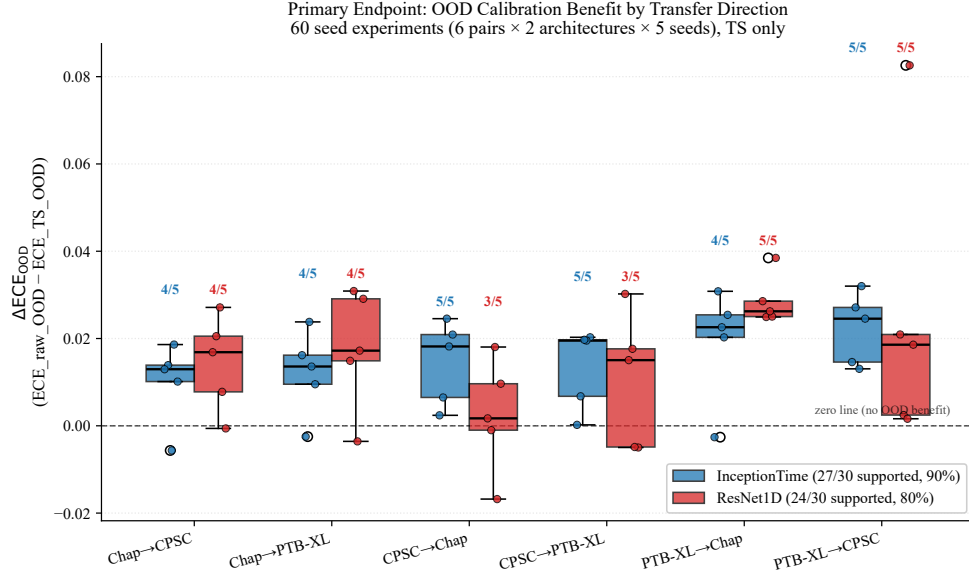


Figure 2: Primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The nine counter-examples are explained in Sec. 6.1.

Stratified report by architecture. The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is qualified, the OOD benefit being about $1.9\times$ the ID

benefit rather than an exclusively OOD effect. The positive ID estimates reflect finite-sample calibration-split variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model selection on the endpoint family), all disclosed.

5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on InceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed coverage on all 6 directions (Figure 4).

Architecture variability. Across the 30 TS experiments per architecture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, including an unexcluded favorable outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion

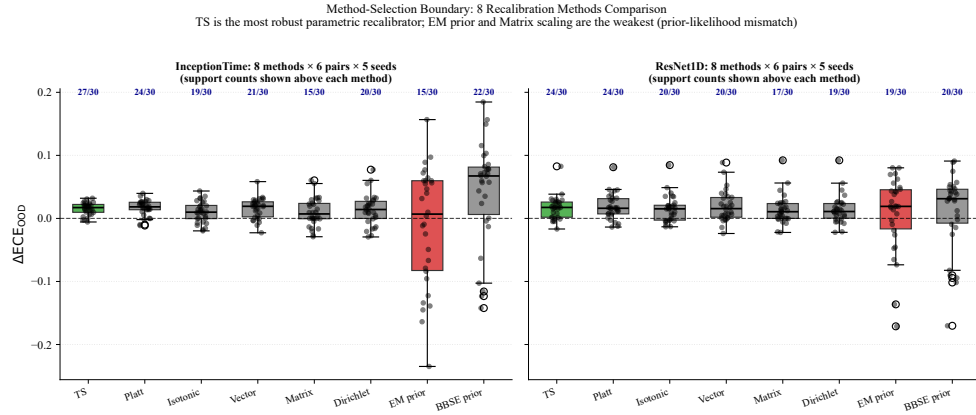


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

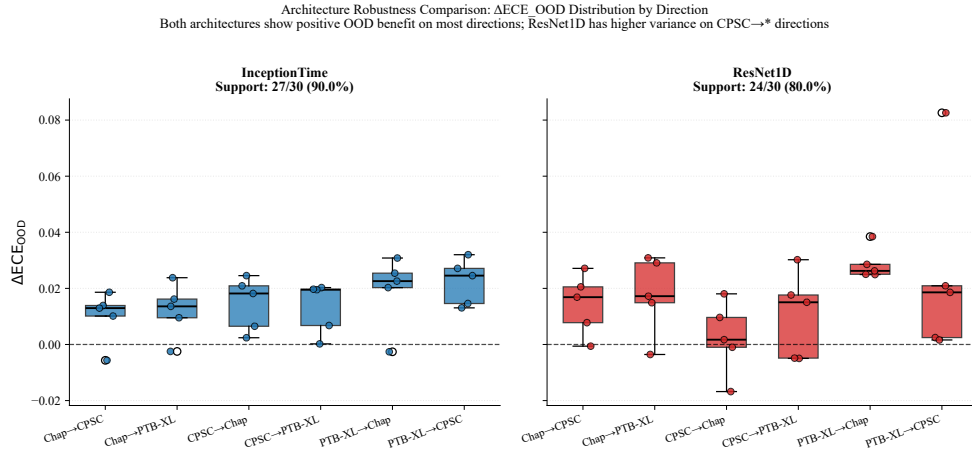


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 15/30 support on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the recovery threshold ($\text{MAPE} < 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20,000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

Honesty caveat (protocol §8). The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE; its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding

clause, “mechanism” wording is limited to “mechanism decomposition (synthetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), versus the registered pair-only folds (6); under that registered deviation the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed “empirical lookup table” rather than “predictive criterion”. No post-hoc predictability claim is made.

Sensitivity analysis with new features. Re-running the LOO R^2 with three label-free features (fitted temperature T , Wasserstein-1 distance between source and target logit distributions, class-wise entropy difference), all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$, entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$ above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8,

based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the appropriate actionable criterion.

5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict “deploy TS” when raw ECE exceeds a threshold, and report sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS $\text{SmoothECE} \leq 0.05$ and ΔECE CI lower bound > 0 , primary-endpoint convention). Comparators: always-none, always-TS, always-buy- n -labels.

Sensitivity / specificity (Table 6): Youden’s J is maximized at raw $\text{ECE} = 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is high (92% of beneficial cases caught) but specificity low (20% of non-beneficial cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure: missing a beneficial recalibration costs more than an unnecessary one. *Caveat:* the threshold is searched on the same 1230-cell pool used for reporting (in-sample). A leave-shift-type-out recomputation (threshold selected excluding the held-out dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (`results/deployment_holdout_recompute.csv`). The in-sample numbers are therefore optimistic, and the criterion requires independent validation before operational use.

Deployment-time measurability. Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap, per-cell labels), requiring target ground-truth labels unavailable in the unlabeled stream. The only label-

free proxies examined are the predictability experiment’s target statistics (predicted-prior L_1 distance, logit moments); that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2). A deployable gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix covers 157 of the 390 planned (pair, seed, shift) combinations (12 complete pair–seed combinations $\times$ 13 levels, plus one partial seed-44 cell at fs250); seeds 42–43 cover all 6 InceptionTime pairs, 44–46 are pending, and ResNet1D L2 is excluded, so these numbers rest on 2-seed, single-architecture coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

Decision-curve analysis. At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (-0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives $+0.014$ – $+0.035$, $T = 1.5$ gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across our 60 seed experiments ranges 0.06 to 0.35 (mean 0.15), 4 – $13\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08 – 0.24), confirming the negative ΔECE reflects genuine TS-induced harm, not metric noise. For large $T > 3$ (noise floor 0.10 – 0.16) the floor is comparable to the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot be excluded there; however, its *direction* (TS increases ECE under perfect calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in $51/60$ experiments, so it cannot explain the positive findings. The primary endpoint is therefore

a real effect, not a SmoothECE artifact, with the caveat that for large- T experiments the noise floor is non-negligible.

5.8. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-class accuracy.

Across 14 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures, PTB-XL $\rightarrow$ Chapman having 3 including Mamba), 7/14 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC, CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (InceptionTime only). The other 7/14 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded too far to be clinically actionable.

This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2 rejects). It supplements C1 by also asking whether discrimination is sufficient for the recalibrated predictions to be clinically useful.

5.9. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error +0.29, median +0.29). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors cluster in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 distance between source and target logit distributions is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting the linear approximation breaks down once the shift invalidates its underlying local Taylor expansion.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid *predictor* (*how much*). For clinical translation, the gate in §5.8, using only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.10. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027 and range $[-0.001, +0.017]$. The directions with the largest net benefit

are CPSC→PTB-XL (mean +0.011) and PTB-XL→CPSC (mean +0.003), which also have the largest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.254 and +0.260 respectively). The direction with negative net benefit is PTB-XL→Chapman (mean −0.005), which is also the direction with the smallest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.004) and is correctly rejected by the discrimination gate (C5). This confirms that the calibration benefit translates to clinical decision benefit in the directions where the gate permits deployment.

5.11. Net Clinical Value (NCV) results

The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across the two secondary endpoints (DCR, NCV).

Result: the OOD NCV is +0.008125 with 95% CI [+0.0014, +0.0148] (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS yields a net improvement in OOD clinical decisions. The ID NCV is −0.006292 with 95% CI [−0.0114, −0.0012] ($p = 0.016$), significantly *negative*: TS slightly worsens ID decisions, consistent with the ID boundary condition (TS compresses confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

5.12. Reliability Diagrams

Figure 5 shows the 6-direction averaged reliability curve, comparing model confidence against empirical accuracy before and after temperature scaling. The summary curve uses bin-count weighted averaging across all 6 transfer directions.

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

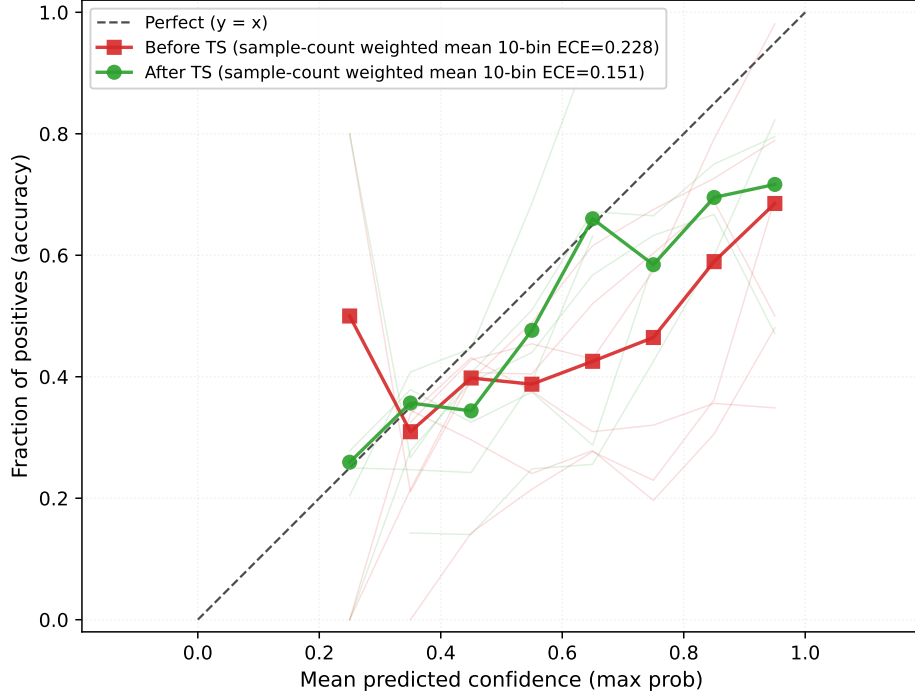


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

Figures 6–8 show the per-direction reliability diagrams for the representative configuration (ResNet1D, seed 42). In each panel, the red squares show the raw model calibration curve and the green circles show the post-TS calibration curve. The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 resamples).

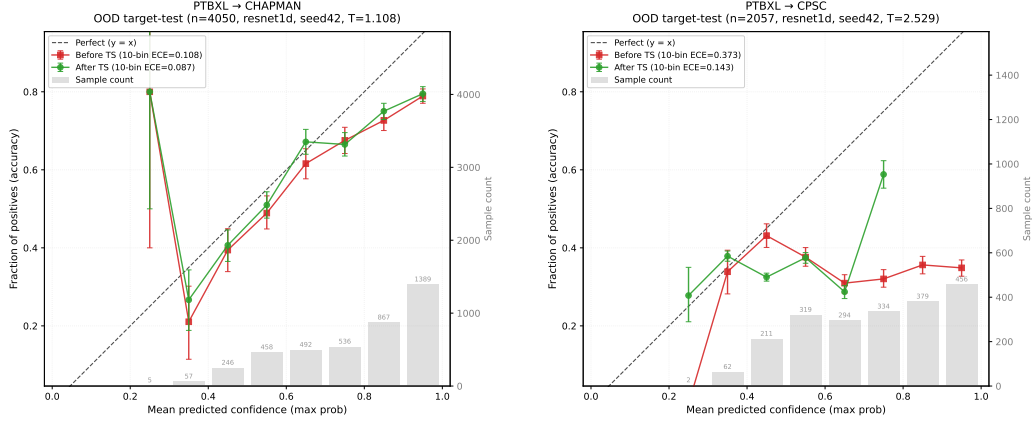


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently disclose all nine rather than excluding them, because the protocol requires all seed results to be reported and each has an identifiable post-hoc mechanism supporting the qualified claim.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature mismatch, not room shortage); and negative decay (-0.0075),

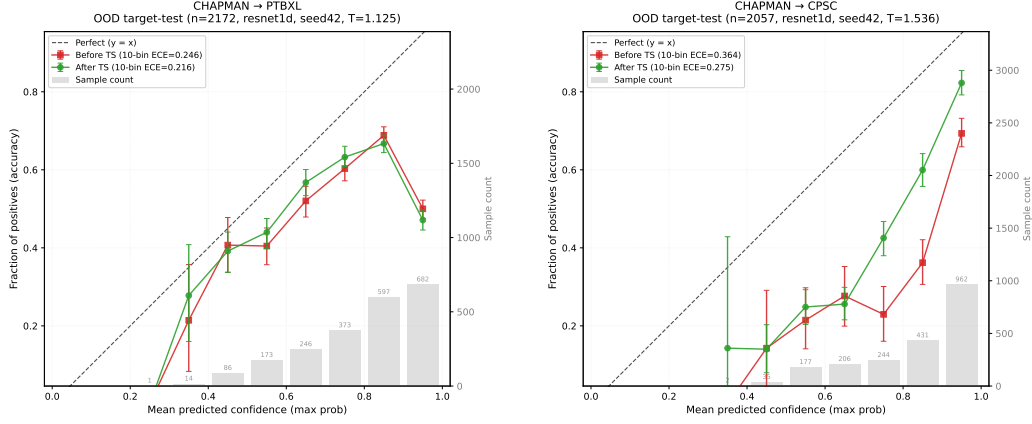


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and

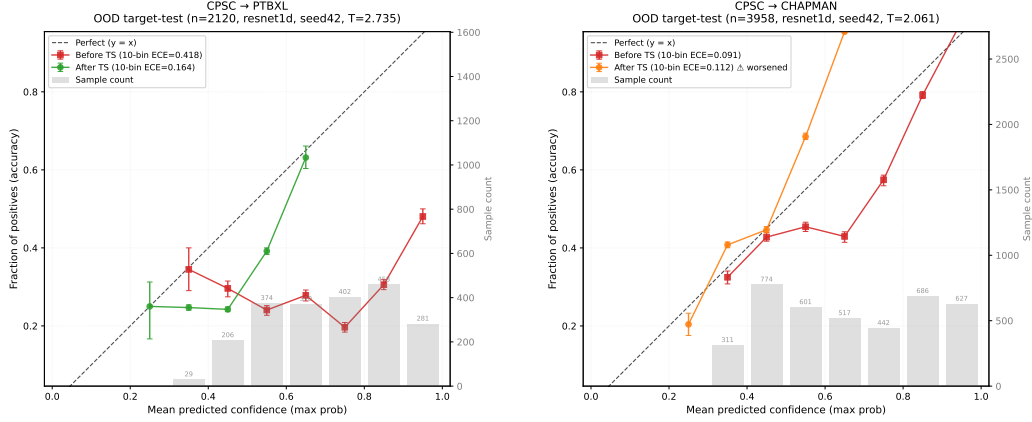


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

$\Delta\text{ECE}_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.

$\Delta\text{ECE}_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta\text{ECE}_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44.

$\Delta\text{ECE}_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the “low OOD accuracy + large gap” mechanism of the other 8.

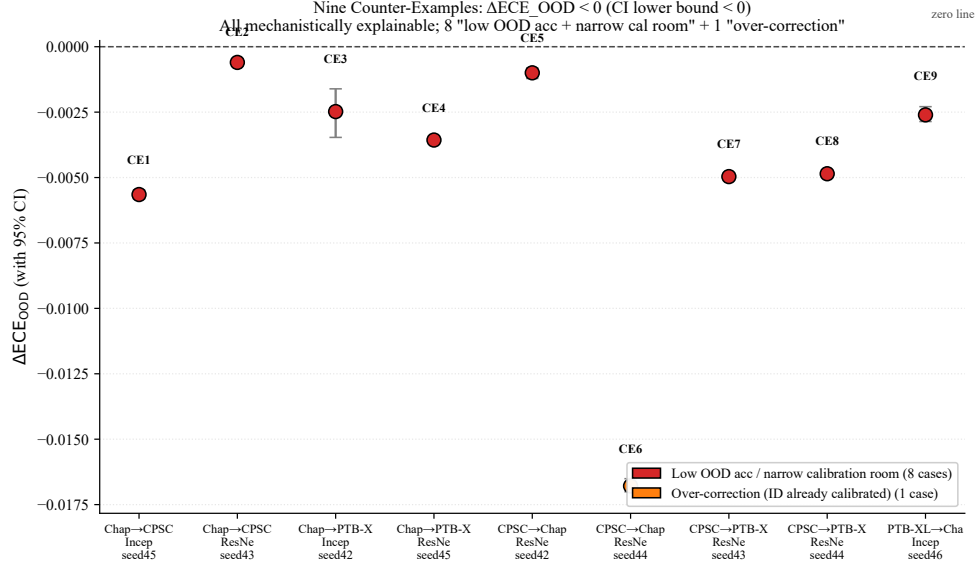


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43.
 $\Delta\text{ECE}_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44.
 $\Delta\text{ECE}_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temperature mismatch; the ninth (CE6) is over-correction on an already well-calibrated backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

Methodological positioning of the 15.0% counter-example rate.
The 15.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All 9 counter-examples have iden-

tifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods achieve higher mean $\Delta \text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763 , BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant positive ID benefit (mean +0.0083 vs. OOD +0.0159; stratum means -0.0008 to $+0.0232$, Table 2). The boundary condition does *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID

benefit is on average about half the OOD benefit, so part of the apparent OOD gain is shared with the ID domain. A val-ECE early-stopping effect (model selection within the endpoint family) plausibly contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively OOD effect. The ID-vs-OOD asymmetry is therefore a *difference in magnitude* between two positive benefits, not presence versus absence.

6.4. Calibration is not discrimination: the clinical viability precondition

Temperature scaling is a monotone transform of the softmax probabilities; it preserves the argmax by construction (asserted bitwise in the released sanity check), so it cannot change classification accuracy, AUROC, or AUPRC. The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes an already-deployable model *trustworthy*; it cannot make a non-discriminative model deployable.

The accuracy statistics make this concrete. Across the 60 TS experiments, mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean OOD accuracy is *below* baseline in 4 of 6 directions: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411 (essentially at baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). There recalibration is moot, since no temperature improves a model that should be replaced by a majority-class predictor, con-

sistent with the counter-example pattern (6/9 with low OOD accuracy). Per-direction AUROC/AUPRC are not computed in the released artifacts (accuracy only; a registered limitation, see Limitations (xiv)), so the viability screen is accuracy-based; the discrimination-first gate precedes the calibration decision rule in deployment.

6.5. *Explainable $\neq$ predictable: post-hoc vs. pre-registered*

We distinguish *explainable* from *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

The counter-examples are explainable but not predictable: the unsupervised target-domain statistics available at deployment time cannot flag them in advance, which is why the predictability experiment triggers the failure branch despite mechanistic explainability. The two-step chain (decomposition + deployment lookup table) provides post-hoc explanation and actionable guidance without a pre-registered predictability claim.

6.6. Transparent re-location vs. HARKing

The pre-registration revision A1 (primary endpoint re-location from **decay** to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [26, 25] is whether the revision direction is theory-driven or result-driven. Our revision direction (“OOD is the calibration benefit source”) is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovdia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. Net clinical value: implications of the directional NCV asymmetry

The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV = +0.008125 ($p = 0.019$) is a small but significant positive value (recalibration improves decisions on unseen target data more often than it degrades them); ID NCV = -0.006292 ($p = 0.016$) is a small but significant negative value (a minor decision-cost penalty on data the source model already calibrates well).

Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60 positive) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/14 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautionary for ID deployment*. It also informs the meta-analytic pooling (Sec. 7),

where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-step); the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.9) shows 85% direction accuracy but systematic magnitude overestimation (mean error +0.29), and a sensitivity analysis with T , Wasserstein-1, and entropy features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses binned ECE, not the main endpoint’s SmoothECE, so quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%) are transparently disclosed, with a frequent post-hoc pattern (low OOD accuracy, non-positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 successful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 experiments are retained as historical provenance (5 lack logs); (viii) the Youden threshold is fitted in-sample (the protocol’s held-out dose levels were not used), making the operating characteristic optimistic; (ix) the L2 deploy-

ment/safety analysis covers 157 of 390 planned (pair, seed, shift) cells at 2 seeds on one architecture, and the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavailable under PhysioNet licensing); (x) temperature was fitted once per experiment, so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap is implemented but not enabled); (xi) model selection used validation ECE rather than the pre-registered NLL early-stopping rule, plausibly contributing to the positive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-probability, and concept-shift components is not asserted, since each pair mixes all three, isolated at eval time; (xiii) the deployment lookup table requires labeled target data, and the label-free proxies failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC, so the viability precondition is screened by accuracy alone, with OOD accuracy below the majority-class baseline in 4 of 6 directions; (xv) NCV is directionally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = -0.006292, $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$ to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits, with 48 as future work; (xvii) this work is from a single-author study, limiting independent review and audit, partially mitigated by the pre-registration protocol and public repository. **Future work:** (1) prospective multi-device validation; (2) continuous shift spaces; (3) π -independent recovery estimators (BBSE/EM-class) to upgrade the MAPE_{π} to a real estimator test; (4) the ResNet1D deficit on CPSC→Chapman and

CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) confirmatory testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6) re-running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU; (7) a multi-threshold permutation test on NCV.

7. Conclusion

We pre-registered and validated a multi-seed calibration boundary study of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile: 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch; the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60), EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/14 strata pass both layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination failures. The predictability failure branch is triggered (three-step chain becomes two-step), and all hypotheses are exploratory + robustness validation, not confirmatory; protocol v2.1-A1 and the five-item honesty declaration (§7) are archived for transparency. Nine counter-examples are transparently disclosed, mostly with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD gap (≥ 0.18 in 7/9), a tendency rather than a sufficient gate (three have OOD accuracy > 0.5). TS needs local validation (safety rate 0.0064),

the Youden criterion is fitted in-sample, and OOD accuracy is below the majority-class baseline in 4 of 6 directions, where TS cannot help. A noise floor control (§5.7) rules out a SmoothECE artifact: worst-case metric noise (+0.021) is 4–13× below the raw ECE (0.06–0.35). The C5 gate (§5.8) rejects every clinically moot stratum. Real-data Shapley cross-fitting (§5.9) gives 85% direction accuracy with magnitude overestimation (mean error +0.29); T , Wasserstein-1, and entropy features confirm the boundary (all $R^2 < 0$). **Meta-analytic pooling.** Both the fixed-effect and DerSimonian–Laird random-effects estimates are tabulated in `meta_analytic_pooled.csv`. The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative (−0.000573) because narrow-CI studies dominate, while DerSimonian–Laird random effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossArchitecture +0.0148; unweighted +0.0152, +0.0165, +0.0159). This reflects heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k - 1$), not a negative central effect, so the count-based proportion (51/60 = 85.0%) is primary; τ^2 varies by 41% (pooled estimate stable within 3.3%). **Family size.** The 60 experiments form the robustness-evidence family; the confirmatory family is the $6 \times 2 \times 13 = 156$ shift grid, and the $8 \times 6 \times 5 \times 2 = 480$ -cell method design is in Sec. 5.3. BH-60 and Bonferroni leave the headline unchanged (51/60); stratum aggregation (12 strata) retains 9/12 directions (InceptionTime 6/6), and Bonferroni-12 retains 2/12 (cpsc_chapman/IT, ptbxl_chapman/RN). **CI width.** Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60 (75.0%; Inception-

Time 24/30, ResNet1D 21/30). Narrow CIs reflect the moderate calibration sample ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 patients) and low SmoothECE variance ($B = 10,000$, CI method=BCa); they do not affect the sign-based support/counter-example classification.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test; independent π recovery (BBSE/EM) is a deferred item (protocol §13).
2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.
4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13). The wording is limited to “mechanism decomposition (synthetic validation part)”.
5. The repository’s `scripts/preprocess_cinc2021.py`, `run_paper_benchmarks.py`, `resources/`, and `assets/` come from the third-party SafeECGMatch release (original README in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment grid.

Data availability

The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019 datasets are publicly available via PhysioNet/Kaggle under their respective licenses. The pre-registered protocol v2.1-A1, its hash manifest (`docs/osf_archive_manifest.json`), the revision A1 document, and the full per-experiment result files (`checkpoints/transfer/**/transfer_result.json`, `results/*.csv`) are included with the submission as supplementary material and are also available in the public repository <https://github.com/gt17641001169-design/ecg-calibration-transportability>.

Code availability

The full pipeline code, preprocessing (reproducible from the public source databases), patient-level split generation, model training, the calibration-method implementations, and all statistical analyses, together with the environment lock file, the patient-level split indices for all five dataset variants and five random seeds, and selected per-experiment result artifacts, is publicly available at <https://github.com/gt17641001169-design/ecg-calibration-transportability>. Trained model checkpoints for the 60-experiment transfer grid are provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance; for double-blind review, an anonymized mirror can be provided as supplementary material.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a general-purpose large language model assistant for language editing and for drafting and refactoring parts of the analysis code. The author reviewed and verified all resulting content and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Toni Guan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Data statement

This study used only publicly available databases: PTB-XL (v1.0.3, PhysioNet), Chapman–Shaoxing (PhysioNet), and CPSC2018+2019 (PhysioNet Challenge). No new raw waveforms were collected. The patient-level split indices generated in this work, the label-mapping tables, the analysis code, and the per-experiment result artifacts are openly available at <https://github.com/gt17641001169-design/ecg-calibration-transportability>; trained model checkpoints are provided as release assets of the same repository.

Declaration of competing interest

The author declares no competing financial or personal interests.

Supplementary Materials

This section catalogs the 60 per-seed reliability diagrams referenced in the main text. Each figure shows the calibration curve, reliability diagram, and ΔECE bar plot for one of the 60 seed experiments (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds). Figures are named `fig_reliability_supp_{pair}_{arch}_seed{s}.pdf` and are included as external files with the submission.

S1. Per-seed reliability diagrams

The 60 figures are organized by transfer pair and architecture. Each figure corresponds to one row in the master results table and provides the visual calibration evidence underlying the ΔECE values reported in the main text. For each direction, ResNet1D and InceptionTime reliability diagrams are shown across seeds 42–46.

S1.1. Chapman→CPSC

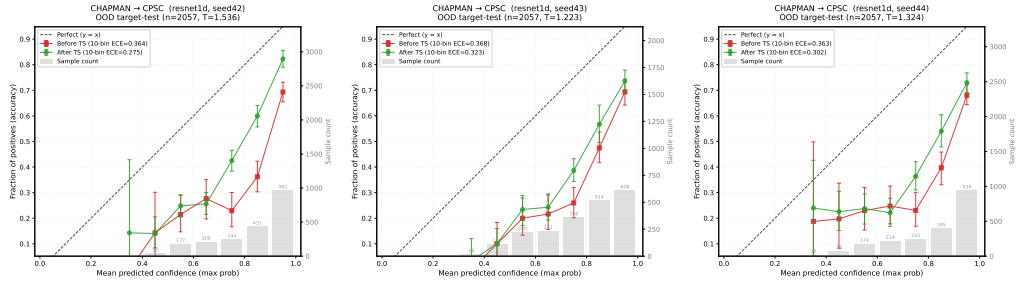


Figure 10: Chapman→CPSC reliability diagrams: ResNet1D seeds 42–44.

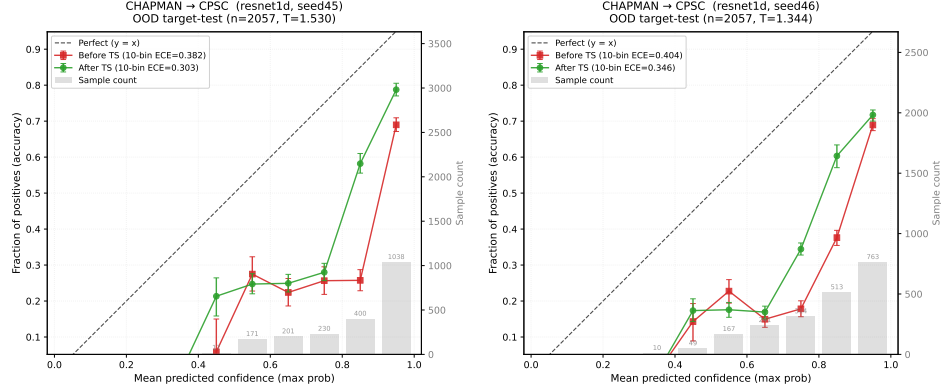


Figure 11: Chapman→CPSC reliability diagrams: ResNet1D seeds 45–46.

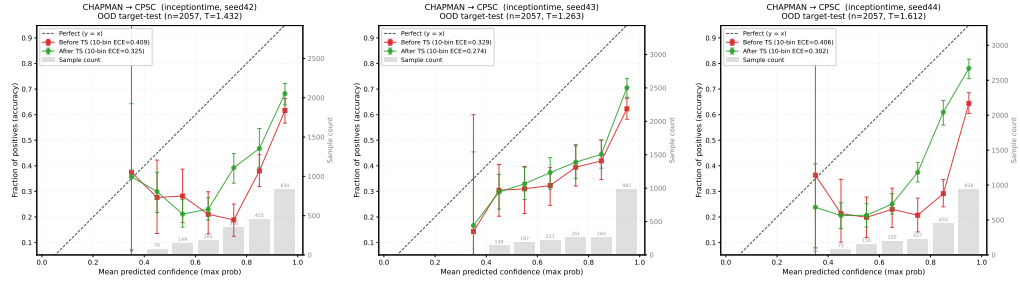


Figure 12: Chapman→CPSC reliability diagrams: InceptionTime seeds 42–44.

S1.2. Chapman→PTB-XL

S1.3. CPSC→Chapman

S1.4. CPSC→PTB-XL

S1.5. PTB-XL→Chapman

S1.6. PTB-XL→CPSC

S2. Additional supplementary materials

- **Per-experiment result files:** full JSON results (checkpoints/transfer/**/transfer_result.json) and CSV summaries (results/*.csv) for all 60 seed experiments.

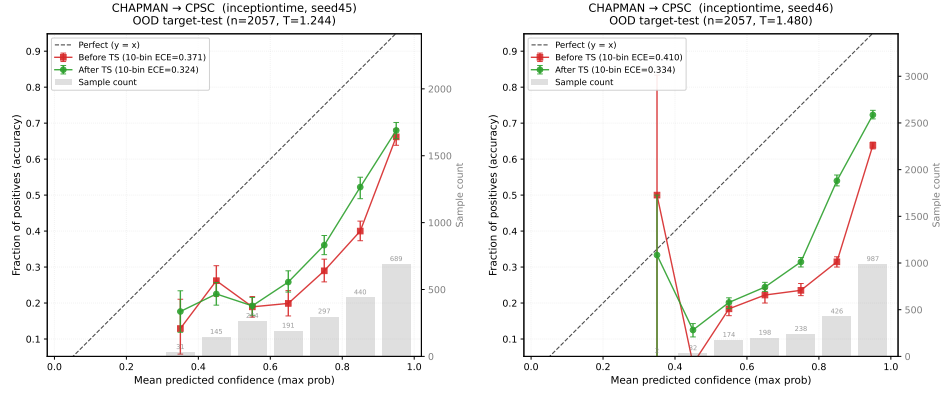


Figure 13: Chapman→CPSC reliability diagrams: InceptionTime seeds 45–46.

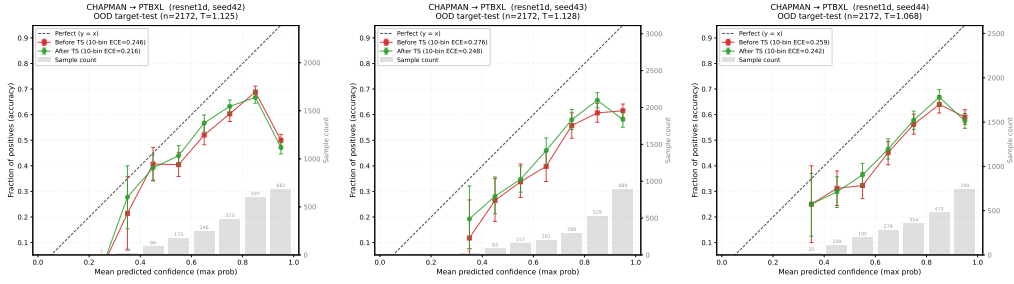


Figure 14: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

- **Pre-registered protocol:** revision A1 full text (docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md) and hash manifest (docs/osf_archive_manifest.json).
- **NCV experiment data:** E3 CSV containing `ncv_id` and `ncv_ood` columns for the net clinical value experiment (Section 4.8).
- **Shapley decomposition cells:** 27-cell validation table for the Shapley attribution analysis at $n = 20,000$.

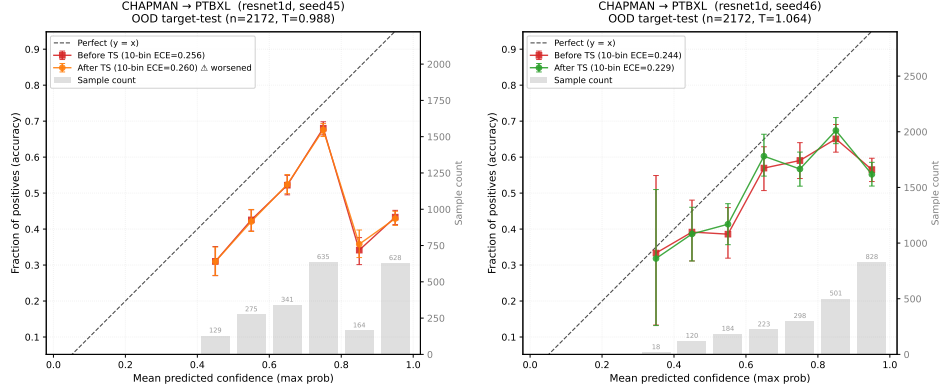


Figure 15: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

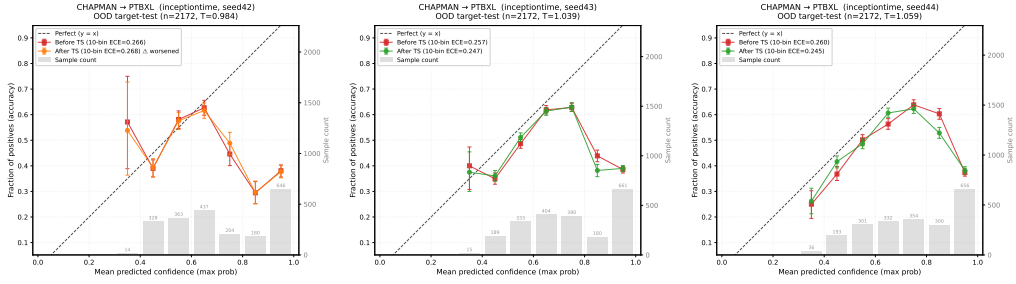


Figure 16: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

- **BCa CI rerun manifest:** full BCa bootstrap ($B = 10,000$) CI results for the entire $6 \times 2 \times 5 \times 8$ grid, stored in the `methods.ts.meta` field of each `transfer_result.json`.

References

- [1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, J. Snoek, “Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift,” *NeurIPS*, 2019. arXiv:1906.02530.

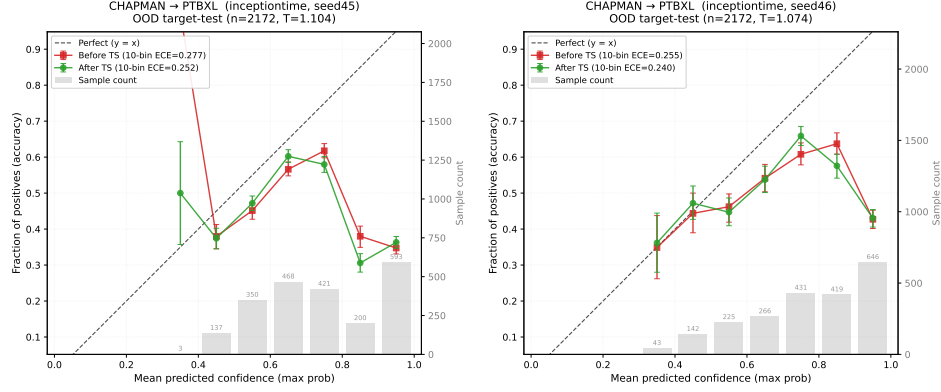


Figure 17: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

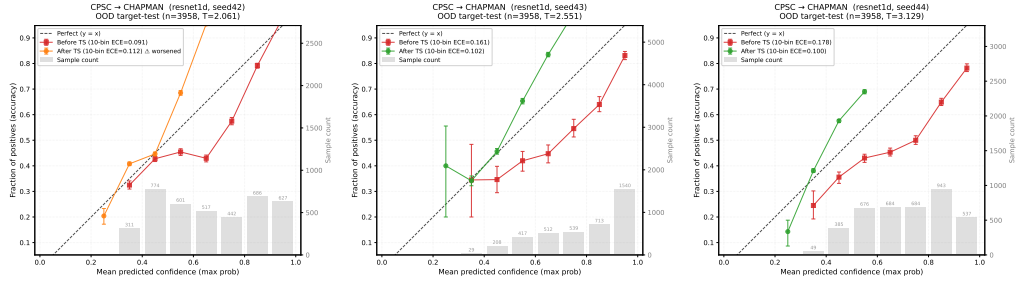


Figure 18: CPSC→Chapman reliability diagrams: ResNet1D seeds 42–44.

- [2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, “On calibration of modern neural networks,” *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.
- [3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., “PTB-XL, a large publicly available electrocardiography dataset,” *Scientific Data*, 7:154, 2020. doi:10.1038/s41597-020-0495-6.
- [4] J. Zheng, J. Guo, H. Chu, “A large scale 12-lead electrocardiogram database for arrhythmia study (Chapman-Shaoxing e-cardiology),” version 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

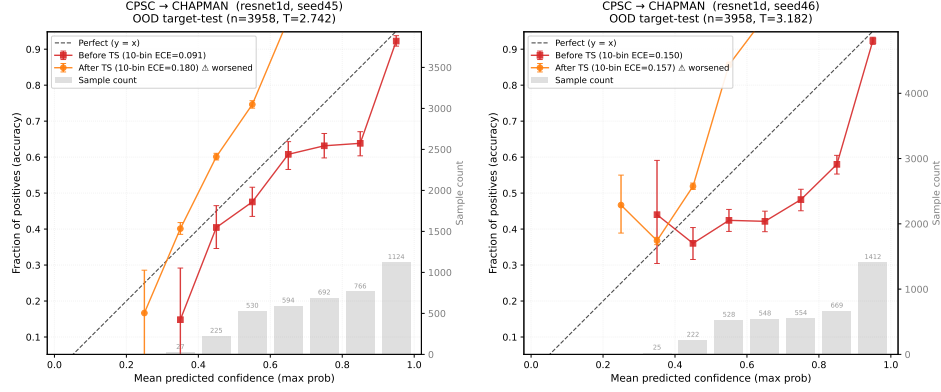


Figure 19: CPSC→Chapman reliability diagrams: ResNet1D seeds 45–46.

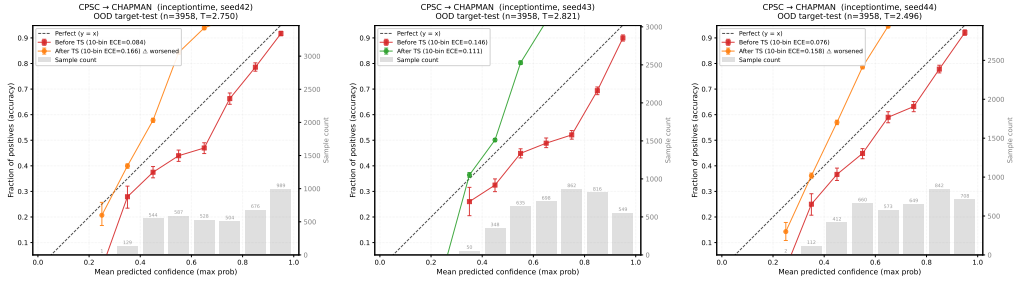


Figure 20: CPSC→Chapman reliability diagrams: InceptionTime seeds 42–44.

- [5] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals,” *Circulation*, 101(23):e215–e220, 2000.
- [6] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.
- [7] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach, “Beyond temperature scaling: obtaining well-calibrated multi-class probabilities with Dirichlet calibration,” *NeurIPS*, 2019.

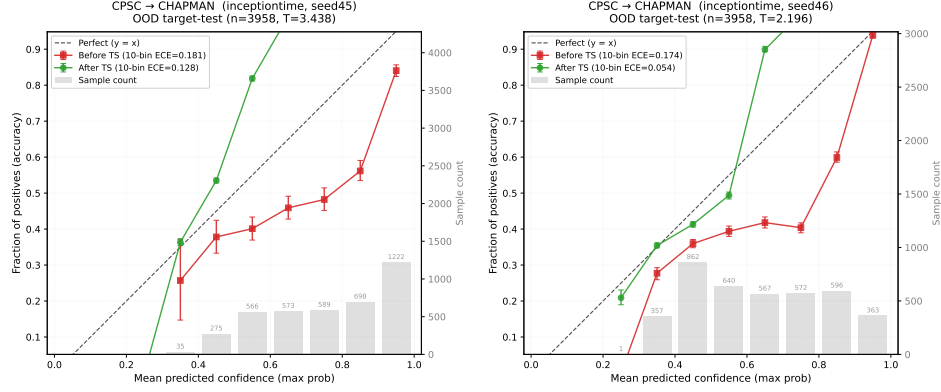


Figure 21: CPSC→Chapman reliability diagrams: InceptionTime seeds 45–46.

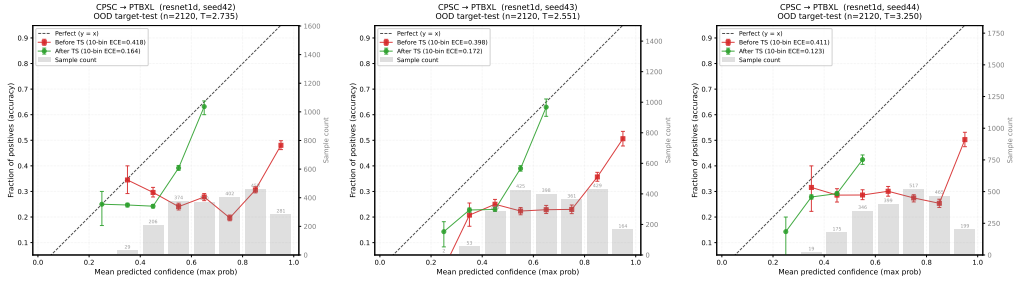


Figure 22: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

arXiv:1910.12656.

- [8] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*, PMLR 119:222–232, 2020. arXiv:1901.06852.
- [9] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018. arXiv:1802.03916.
- [10] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão,

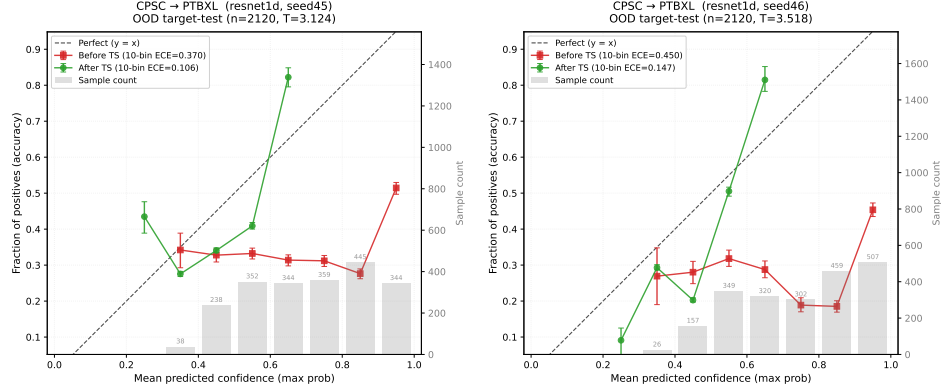


Figure 23: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

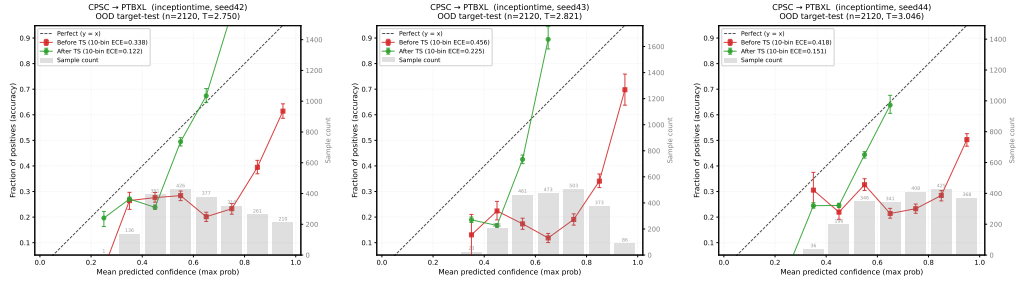


Figure 24: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods in multi-label classification: a case study with automatic diagnosis of electrocardiogram,” *Information Fusion*, 101:101978, 2024. doi:10.1016/j.inffus.2023.101978.

- [11] J. F. Vranken et al., “Uncertainty estimation for deep learning-based automated analysis of 12-lead electrocardiograms,” *European Heart Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.

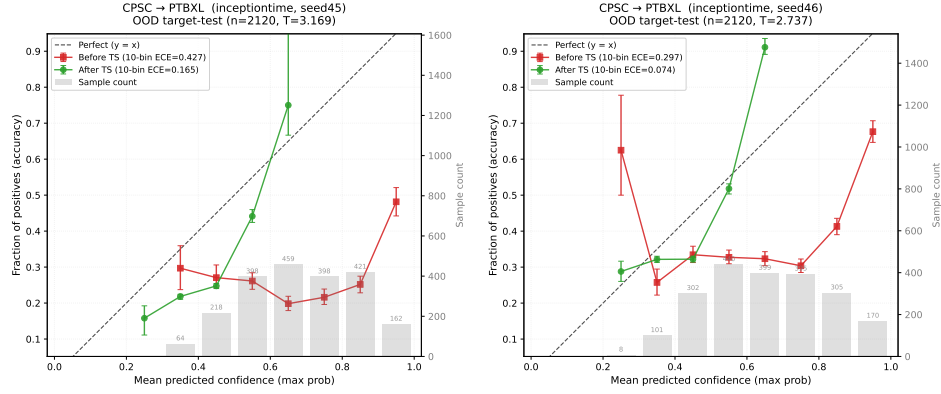


Figure 25: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

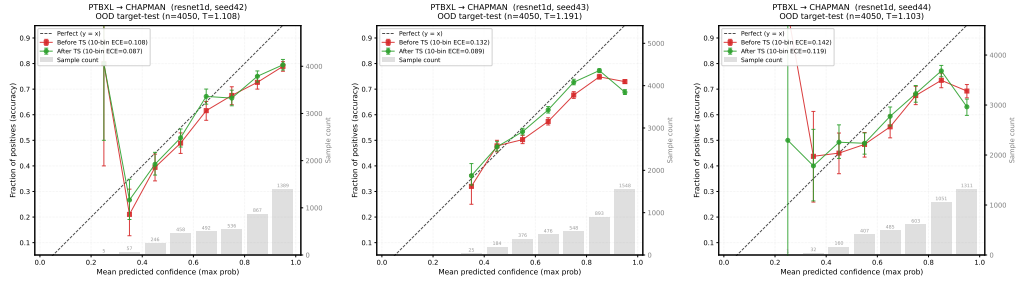


Figure 26: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 42–44.

- [12] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated classification of multilead ECGs using heterogeneous labels and populations,” *Physiological Measurement*, 43(8):084001, 2022.
doi:10.1088/1361-6579/ac79fd.
- [13] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American Statistical Association*, 82(397):171–185, 1987.
doi:10.1080/01621459.1987.10478410.
- [14] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer,

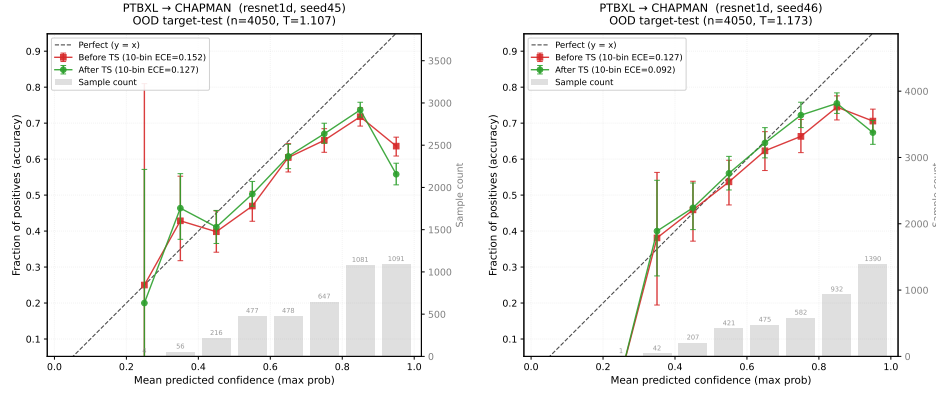


Figure 27: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 45–46.

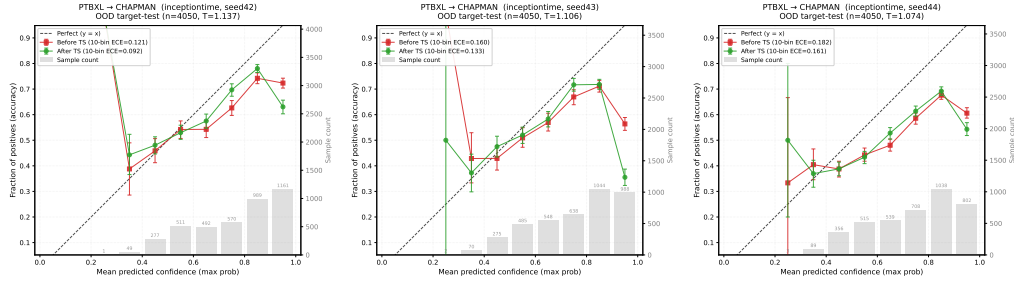


Figure 28: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 42–44.

“A calibration hierarchy for risk models was defined: from utopia to empirical data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016. doi:10.1016/j.jclinepi.2015.12.005.

- [15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins, “Calibration: the Achilles heel of predictive analytics,” *BMC Medicine*, 17:230, 2019. doi:10.1186/s12916-019-1466-7.
- [16] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,” *NeurIPS*, 2019. arXiv:1909.10155.

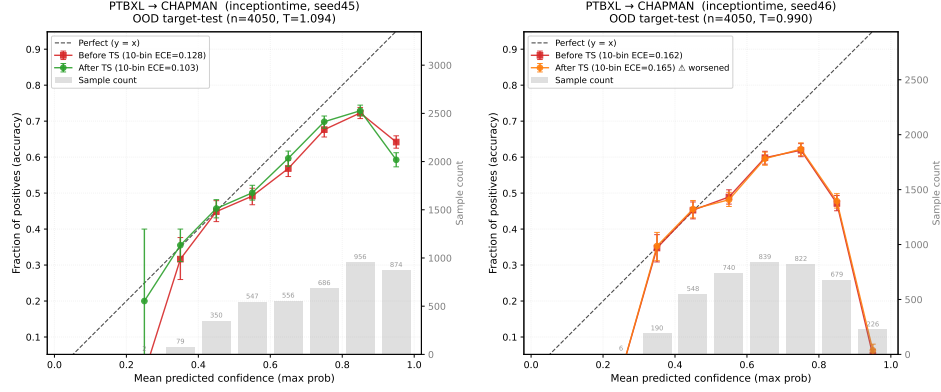


Figure 29: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 45–46.

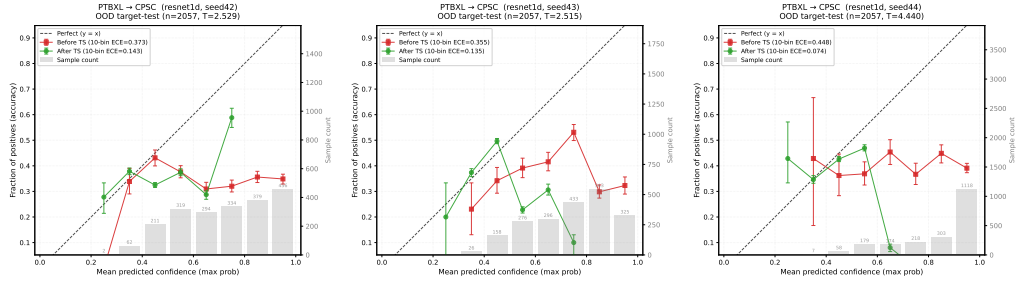


Figure 30: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 42–44.

- [17] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for ECG analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. doi:10.1109/JBHI.2020.3022989.
- [18] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto, “RR-interval-based atrial fibrillation detection and burden estimation: cross-dataset validation and calibration-aware probability analysis,” *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.

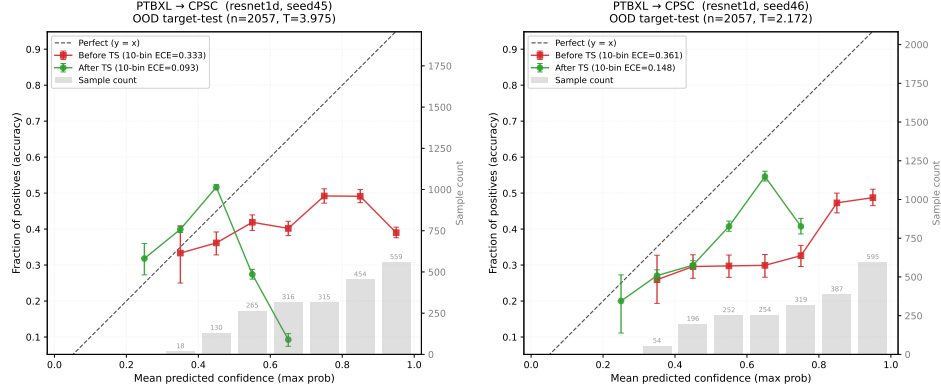


Figure 31: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 45–46.

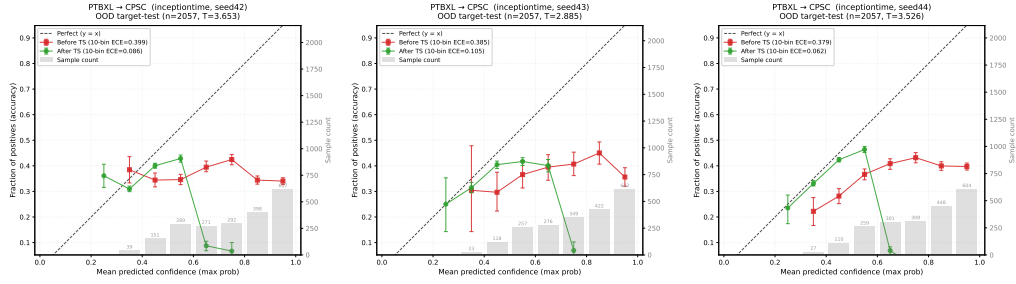


Figure 32: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 42–44.

- [19] K. Patel, P. Beedala, “Calibration drift under cross-institutional deployment: an external validation framework for ICU mortality prediction across MIMIC-IV and eICU,” *medRxiv* (preprint), 2026. doi:10.64898/2026.05.03.26352335.
- [20] X. Wang, M. Long, J. Wang, M. I. Jordan, “TransCal: transferability estimation for auto-selecting transfer calibration,” *NeurIPS*, 2020. arXiv:2007.08259.
- [21] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, “A unified view of label

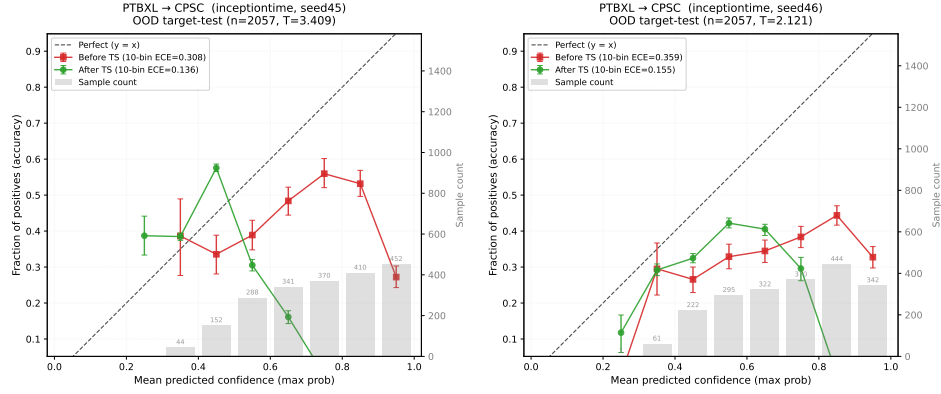


Figure 33: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 45–46.

shift estimation,” *NeurIPS*, 2020. arXiv:2003.07554.

- [22] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., “LaSCal: label-shift calibration for uncertainty quantification,” *NeurIPS*, 2024.
- [23] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera, “A unifying view on dataset shift in classification,” *Pattern Recognition*, 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.
- [24] [Unpublished pre-registered protocol] “ECG calibration boundary study – pre-registered experiment protocol v2.1-A1,” 2026. Hash manifest: docs/osf_archive_manifest.json (local snapshot; public OSF archival pending); revision A1 full text in docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md.
- [25] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, “The preregistration revolution,” *PNAS*, 115(11):2600–2606, 2018. doi:10.1073/pnas.1708274114.

- [26] D. Lakens, “The practical alternative to the p value is the correctly used p value,” *Perspectives on Psychological Science*, 16(3):639–648, 2021. doi:10.1177/1745691620958012.
- [27] J. Li, A. D. Aguirre, V. M. Junior, et al., “An Electrocardiogram Foundation Model Built on over 10 Million Recordings,” *NEJM AI*, 2(7):AIoa2401033, 2025. doi:10.1056/AIoa2401033.
- [28] J. C. McHugh, B. Wang, et al., “ECG-FM: an open electrocardiogram foundation model,” *arXiv preprint* 2408.05178, 2024.
- [29] J. Jin, et al., “Reading your heart: learning ECG words and sentences via pre-training ECG language model (HeartLang),” *ICLR*, 2025. arXiv:2502.10707.
- [30] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it effective?” *PLOS ONE*, 20(5):e0316043, 2025. doi:10.1371/journal.pone.0316043.
- [31] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164, 2026.
- [32] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347, 2026 (accepted at MICCAI 2026).
- [33] N. Nemati, “Comparative analysis of data augmentation for clinical ECG classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv preprint* 2510.24740, 2025.

- [34] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment,” *arXiv preprint* 2605.00647, 2026.
- [35] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 /	0.216 → 0.200	0.217 → 0.201	raw → TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155